Millets: India's Traditional Superfoods

Introduction to Millets

Millets are a group of small-seeded, hardy cereal grasses that have been grown and eaten for thousands of years, particularly in Asia and Africa. They are among the oldest cultivated crops, and evidence of millet cultivation in India goes back to the Indus Valley civilisation, about 4,500 years ago or more. Millets are rich in fibre, minerals and proteins, grow in poor soils with little rain, and are naturally gluten-free. In India they were the staple food of ordinary people for centuries until the Green Revolution of the 1960s shifted attention to rice and wheat. The government now calls millets Shree Anna, meaning the "mother of all grains", and also uses the term nutri-cereals. India is the world's largest producer of millets, accounting for roughly 40 percent of global production, and the main millet-growing states are Rajasthan, Karnataka, Maharashtra, Uttar Pradesh, Madhya Pradesh, Tamil Nadu and Gujarat.

Name Four Millets

Four important millets are ragi, bajra, jowar and foxtail millet. Ragi, also called finger millet (kezhvaragu in Tamil, nachni in Hindi), is especially important in Karnataka and Tamil Nadu. Bajra is pearl millet (kambu in Tamil), common in Rajasthan and Gujarat. Jowar is sorghum (cholam in Tamil), widely grown in Maharashtra and Karnataka. Foxtail millet (thinai in Tamil, kangni in Hindi) is one of the oldest cultivated millets. Millets are divided into major millets (jowar, bajra and ragi) and minor millets, sometimes called small millets, which include foxtail millet (thinai), little millet (samai), kodo millet (varagu), barnyard millet (kuthiraivali), proso millet (panivaragu) and browntop millet. Traditional Tamil Sangam literature mentions thinai, and the word is also used for the five landscapes of ancient Tamil poetry.

Why Was 2023 Important for Millets?

The year 2023 was declared the International Year of Millets by the United Nations General Assembly. The proposal was made by India and was supported by more than 70 countries, and the resolution was adopted in March 2021. The Food and Agriculture Organization (FAO) of the United Nations led the celebrations. The aim of the year was to raise global awareness of the health benefits of millets and their suitability for farming in a changing climate, to encourage production and consumption, and to improve food security. India also celebrated it with events, millet festivals and a huge push through the Indian Institute of Millets Research in Hyderabad, which was named a Global Centre of Excellence for millets. The government of India had earlier declared 2018 as the National Year of Millets, and in the 2023 Union Budget, millets were given a special focus. Tamil Nadu launched a Millet Mission in 2023 to promote millet cultivation and consumption.

What Are the Health Benefits of Ragi?

Ragi, or finger millet, is one of the most nutritious cereals. It is the richest known plant source of calcium among cereals, with about 340 to 350 milligrams per 100 grams, which is many times more than rice or wheat, so it strengthens bones and teeth and helps prevent osteoporosis. It also contains a good amount of iron, which helps prevent anaemia, particularly when eaten with vitamin C-rich foods; it is high in dietary fibre, which aids digestion and gives a feeling of fullness useful in weight control; and it has plenty of amino acids such as methionine. Ragi has a lower glycaemic index than polished rice, so it releases sugar slowly into the blood and is helpful for people with diabetes. It is gluten-free, suits people with gluten intolerance, and is a traditional weaning food for babies in the form of ragi porridge. It is rich in antioxidants (polyphenols), which protect cells. Popular ragi dishes include ragi koozh or kanji, ragi dosa, ragi mudde (balls) in Karnataka, ragi roti, ragi malt and ragi laddu.

Benefits of Other Millets

Bajra is rich in iron, magnesium and protein, and gives warmth in winter. Jowar provides fibre and B vitamins and is used for rotis. Foxtail millet has a low glycaemic index and is used for upma and pongal. Little millet and kodo millet are rich in fibre and help in managing blood sugar. Barnyard millet has a very high fibre content and is eaten during fasts. Overall, millets help in controlling diabetes, protect the heart by lowering cholesterol, aid digestion and help in weight management. Millets have a coarse texture and may require soaking or fermenting to improve digestion.

Why Millets Are Good for Farmers and the Planet

Millets are climate-resilient crops. They need much less water than rice or sugarcane, often only about one-third to one-half the water of rice, and many varieties mature in 60 to 90 days. They thrive in dry, rainfed regions and poor soils, need few chemical fertilisers and pesticides, resist heat and drought and cause lower greenhouse gas emissions, making them important for farmers in a warming world. Because they are nutritious and hardy, they can help fight malnutrition and anaemia.

Millets in Tamil Nadu and Indian Cuisine

Tamil Nadu has a rich tradition of millet foods such as kambu koozh, kezhvaragu (ragi) kali, thinai pongal, samai upma and varagu rice. Districts such as Dharmapuri, Krishnagiri, Salem and the Nilgiris grow ragi, and Perambalur and Thoothukudi grow cumbu. The state has millet cafes, millet festivals, and supplies millets through the public distribution system in some districts. Rajasthan's bajra roti and Karnataka's ragi mudde are famous elsewhere in India. Millets are included in mid-day meals under the PM POSHAN scheme in some states.

Challenges and Future

Challenges include lower consumer demand, difficult processing such as de-husking small millets, shorter shelf life of millet flour and lower yields compared with rice and wheat. The future includes millet-based snacks, breakfast cereals and baby foods, better processing machines, and increased minimum support prices.

Quick Facts

- Four millets are ragi (finger millet), bajra (pearl millet), jowar (sorghum) and foxtail millet.
- 2023 was declared the International Year of Millets by the United Nations on India's proposal.
- Ragi is rich in calcium, iron and fibre and has a low glycaemic index.
- India is the world's largest producer of millets and calls them Shree Anna.
- Millets are climate-resilient and need much less water than rice.